

1. CORE INSPECTION METADATA

Timestamp	2026-09-23T22:10:26.814139
Primary Finding	DENT
Severity Layer	HIGH
Detections Logged	1
Analysis Engine	YOLOv11-Nano (Local Edge)

2. ACQUIRED IMAGERY & BOUNDING BOXES



Fig 1. AI-annotated structural imagery overlay.

3. RAW DETECTION COORDINATES

ID	DEFECT CLASS	CONFIDENCE	SEVERITY	BOUNDING BOX [X, Y, W, H]
01	DENT	86.0%	HIGH	[0.430, 0.418, 0.310, 0.207]

4. RAG CONTEXT RETRIEVAL

Query N/A

Source Manual ac_43.13-1b_w-chg1.pdf (Page 150)

Extracted Snippet: *"sustain these stresses, and then allow them to return into the structure. The repair must be equal to the original structure, but not stronger or stiffer, which will cause stress concentrations or alter the resonant frequency of the structure. c. All-metal aircraft are made of very thin sheet metal, and it is possible to restore the strength of the skin without restoring its rigidity. All repairs should be made using the same type and thickness of material that was..."*

5. SYNTHESIZED REPAIR PROTOCOL

Generated via Local Phi-3 Mini strictly grounded in retrieved maintenance documentation.

- 1 Isolate the damaged area**
Clearly mark and isolate the high-severity dent on the aircraft's surface to define the extent of the repair needed.
- 2 Remove the dented material**
Carefully remove the dented metal sheet from the aircraft's surface, ensuring not to cause additional damage to the surrounding area.
- 3 Prepare the repair material**
Select the same type and thickness of metal sheet used in the original construction to maintain structural integrity and resonant frequency.
- 4 Perform the repair**
Skillfully apply the prepared metal sheet to the dented area, ensuring a seamless and flush fit without over-tightening, which could introduce new stresses.
- 5 Inspect and test the repair**
Conduct a thorough inspection of the repaired area and perform necessary tests to ensure the repair meets the required specifications and does not alter the aircraft's resonant frequency.